

Beginner to Working RAG in n8n Cloud (Validated Setup Guide)

This guide is based on the exact configuration and troubleshooting steps that were successfully validated during our setup.

1. Solution Overview

Goal: Build a Retrieval-Augmented Generation (RAG) chatbot using n8n Cloud, Qdrant Cloud, Hugging Face Embeddings, and Groq.

Final Architecture:

PDF -> Extract Text -> Chunk Documents -> Generate Embeddings -> Store in Qdrant

User Question -> AI Agent -> Qdrant Retrieval -> Groq -> Answer

Screenshot Placeholder: Insert final workflow screenshot here.

2. Prerequisites

Accounts required:

- n8n Cloud
- Qdrant Cloud
- Hugging Face
- Groq

Validated Embedding Model:

sentence-transformers/all-MiniLM-L6-v2

Vector Dimension: 384

Distance Metric: Cosine

3. Qdrant Cloud Setup

Create a cluster and save:

- Cluster URL
- API Key

Create collection:

Name: documents

Dimensions: 384

Distance: Cosine

Common Error:

Dimension mismatch.

Fix:

Collection dimension must match embedding model dimension (384).

Screenshot Placeholder: Qdrant collection creation screen.

4. Hugging Face Setup

Generate a Hugging Face token.

Read permission was sufficient for our setup.

Configure credential in n8n.

Model used:

sentence-transformers/all-MiniLM-L6-v2

Screenshot Placeholder: Hugging Face token screen.

5. Groq Setup

Create API key in Groq.

Configure Groq Chat Model in n8n.

Recommended models:

- llama-3.3-70b-versatile
- llama-3.1-8b-instant

Screenshot Placeholder: Groq credential screen.

6. Document Ingestion Workflow

Workflow:

Manual Trigger

-> Extract From File

-> Qdrant Vector Store

Connected nodes:

- Hugging Face Embeddings
- Default Data Loader

Screenshot Placeholder: Ingestion workflow.

7. Why Local File Upload Failed

n8n Cloud cannot directly access your local filesystem.

Read/Write Files node is intended for self-hosted deployments.

Working solution:

Use On Form Submission and File Upload.

8. Form Upload Workflow

Trigger:

On Form Submission

Add Form Element:

Label: Upload PDF

Type: File

Upload PDF through generated form URL.

Screenshot Placeholder: Form configuration.

9. Extract From PDF Configuration

Node:

Extract From File

Operation:

Extract From PDF

Important Discovery:

Input Binary Field must be:

Upload_PDF

This value came from the uploaded file field name.

Common Error:

The item has no binary field 'data'

Fix:

Replace 'data' with Upload_PDF.

Screenshot Placeholder: Extract From PDF node.

10. Verifying PDF Extraction

Successful extraction shows:

- numpages
- metadata
- text

Verify extracted text before moving forward.

Screenshot Placeholder: Successful PDF extraction output.

11. Default Data Loader Configuration

Type of Data: JSON

Mode: Load All Input Data

Text Splitting: Simple

Observation:

Default Data Loader already performed chunking successfully.

No custom Recursive Text Splitter was required.

Screenshot Placeholder: Default Data Loader settings.

12. Qdrant Insert Documents

Operation Mode:

Insert Documents

Collection:

documents

Embedding:

sentence-transformers/all-MiniLM-L6-v2

Observed Result:

~35 chunks inserted.

Screenshot Placeholder: Qdrant insertion result.

13. Verifying Ingestion

Signs of success:

- Multiple chunks visible
- pageContent populated
- metadata populated
- no insertion errors

Screenshot Placeholder: Qdrant output table.

14. Chat Workflow

Workflow:

When Chat Message Received

-> AI Agent

Connected Components:

- Groq Chat Model
- Qdrant Vector Store Tool
- Hugging Face Embeddings

Screenshot Placeholder: Chat workflow.

15. Qdrant Retrieval Tool

Operation:

Retrieve Documents (As Tool for AI Agent)

Collection:

documents

Limit:

4

Include Metadata:

Enabled

Screenshot Placeholder: Retrieval configuration.

16. AI Agent Configuration

System Prompt:

Answer only using information retrieved from the documents.

If the answer is not found in the documents, say:
'I could not find that information in the knowledge base.'

Screenshot Placeholder: AI Agent prompt.

17. Validation Tests

Successful Queries:

- What is ANI?
- What problem does ANI solve?
- List the key features of ANI?

These queries successfully retrieved document content.

Screenshot Placeholder: Successful answer output.

18. Troubleshooting Guide

Issue: No binary field data

Fix: Use Upload_PDF

Issue: Empty retrievals

Fix: Verify collection name and embeddings

Issue: Poor answers

Fix:

Increase retrieval limit from 4 to 8-10

Issue: Dimension mismatch

Fix:

Collection dimension must be 384

Issue: Nothing ingested

Fix:

Verify extracted text exists before Qdrant insertion

19. Recommended Enhancements

- Increase retrieval limit to 8-10
- Enable reranking
- Add conversational memory
- Upload multiple PDFs

- Create separate collections per project
- Add metadata filtering

20. Final Rebuild Checklist

1. Create Qdrant cluster
2. Create documents collection
3. Configure Hugging Face embeddings
4. Configure Groq
5. Build ingestion workflow
6. Upload PDF
7. Verify extraction
8. Verify chunking
9. Insert into Qdrant
10. Build chat workflow
11. Configure retrieval tool
12. Test with document-specific questions

Result:

Working RAG chatbot using n8n Cloud + Qdrant + Hugging Face + Groq.